INTERNAL SPECIFICATION FOR **CERAMIC TILES**

Sr. No	CHARACTERISTICS	SF HD 31.8x31.8x1.2 cm		Roofing Tile 31.8x31.8x1.2 cm	
1	Nominal size **(mm)**	318.0x318.0x12.0		318.0x318.0x12.0	
2	Work size **(mm)**	318.5x318.5X12.0		318.5x318.5X12.0	
		ISO 13006	**SCC. spec.**	**ISO 13006**	**SCC. spec.**
3.1	Size variation from work size **(%)**	± 0.60	± 0.60	± 0.60	± 0.60
	In **mm**	± 1.91	± 1.91	± 1.91	± 1.91
3.2	Size range in **mm**	316.6-320.4	316.6-320.4	316.6-320.4	316.6-320.4
3.3	Size code 01	-	316.6-318.0	-	316.6-318.0
	Size code 00 **(Normal)**	-	318.1-319.0	-	318.1-319.0
	Size code 02	-	319.1-320.4	-	319.1-320.4
4	Thickness deviation **(%)**	± 10.0	± 5.0	± 10.0	± 10.0
5	Thickness Range **(mm)**		11.50 - 12.50		11.50 - 12.50
6	Centre Curvature	mm	mm	mm	mm
6.1	a) Convex **(+)**	+ 2.00	+ 1.50	+ 2.00	+ 2.00
6.2	b) Concave **(-)**	- 2.00	- 0.30	- 2.00	- 2.00
7	Warpage	± 2.00	± 1.30	± 2.00	± 2.00
8	Edge Curvature				
8.1	a) Convex **(+)**	+ 1.59	+ 0.90	+ 1.59	+ 1.59
8.2	b) Concave **(-)**	- 1.59	- 0.30	- 1.59	- 1.59
9	Max deviation from Straightness (**mm**)	± 1.50	± 0.80	± 1.50	± 1.50
10	Max deviation from Rectangularity (**mm**)	± 1.59	± 1.30	± 1.59	± 1.59
11	Water Absorption **(%)**	> 10	16 ± 2	> 10	16 ± 2
12	Coefficient to Linear thermal expansion **($x10^{-6}K^{-1}$)**	-	≤ 8	-	≤ 8
13	Resistance to surface abrasion **(Class)**	Manufacturer to state class	2 - 5	Manufacturer to state class	2 - 5
14	Breaking strength **(N)**	≥ 600	≥ 800	≥ 600	≥ 600
15	Modulus of Rupture **(N/mm^2)**	Min. 12	Min 15	Min. 12	Min 12
16	Crazing resistance **(5bar/2 hr cycle)**	Required	Resistant	Required	Resistant
17	Scratch resistance **(Moh's Scale)**	-	Min.3(Glossy) Min.6 (Matt)	-	Min.3(Glossy) Min.6 (Matt)
18	Staining resistance **(Min.Class)**	3	3	3	3
19	Chemical resistance to Acids and alkalis **(Min.Class)**	Required if agreed	B	Required if agreed	B
20	Resistance to household Chemicals **(Min.Class)**	B	B	B	B